Technical file — PixelPine rev A

Kept by the manufacturer for **10 years** after the last unit is placed on the market (EMC Art. 7, RoHS Art. 7, GPSR Art. 9). Market-surveillance authorities (CZ: ČOI, and ČTÚ for EMC) can request it in Czech or English.

Index

#	Document	Source / status
1	General product description, intended use, photos	this file (§ Description) + `hardware/renders/`
2	Schematic	`hardware/pixelpine_schematic.pdf` ✓
3	PCB layout, assembly drawings, BOM with manufacturer part numbers	`hardware/pixelpine.kicad_pcb`, `docs/bom.md`, `hardware/production/` ✓
4	List of standards applied	EN IEC 55015:2019+A11:2020, EN 61547:2009, EN IEC 63000:2018; EN 60598-2-20 and EN 62471 used as state-of-the-art references for the GPSR risk assessment
5	EMC test report	accredited lab, ◔ planned (test-plan.md)
6	EMC assessment (why the product meets the essential requirements, with design measures)	README.md § Design decisions ✓
7	RoHS: supplier declarations for all 8 LCSC parts + PCB (LF-HASL) + XRF screen	◔ collect from LCSC product pages / JLC
8	REACH SVHC statement	◔ supplier declarations
9	GPSR risk assessment	this file (§ Risk assessment) ✓
10	Photometric note: ≈ 2 lm, so outside 2019/2020 and 2019/2015	`docs/power-and-connectivity.md` ✓
11	EU Declaration of Conformity (signed)	declaration-of-conformity.md, ◔ sign after 5 and 7
12	Labelling, packaging and user information	labelling-and-instructions.md ✓
13	Production control: programming + functional test procedure, batch numbering scheme	this file (§ Production control) ✓
14	Firmware version identification	firmware version readable over SWD; version stored on the batch record

Description

A 70 mm tall, 1.6 mm FR-4 printed circuit board in the shape of a Christmas tree, with 240 warm-white 0603 LEDs, a PY32F003 microcontroller and a USB-C power-only receptacle. It stands on a slotted FR-4 stand fin. It is powered from any USB power source at 5 V and draws at most 19 mA (0.1 W). A capacitive touch electrode shaped as a star (on both faces of the board) lets the user interact with the tree's animated character. Indoor decorative use only. No battery, no radio, no mains connection.

Risk assessment (GPSR Art. 6 & 9)

Method: hazard → scenario → severity (1–4) × probability (1–5) → risk. Anything ≥ 8 must be reduced.

Hazard	Scenario	S	P	Risk	Measures	Residual
Electric shock	User touches the circuitry	1	1	1	SELV 5 V from a CE USB adapter; no mains inside	1
Overheating / fire	Short circuit from a metallic object on the board	2	2	4	Max 0.1 W normally. A USB source current-limits a short (USB 2.0: 500 mA+); FR-4 is UL94 V-0. Leaflet: "don't leave near flammable decorations while unattended" — conservative wording	2
Overheating	Faulty/cheap USB adapter delivering > 5.5 V	2	2	4	MCU abs. max 6.5 V; failure mode is a dead MCU, not fire (currents limited by 100 Ω line resistors). Leaflet: use a CE-marked adapter	2
Cuts / pokes	Sharp PCB tips	2	2	4	**Rounded outline** (min radius 0.6 mm at inner joins, 2.4 mm at tips), routed edges	1
Choking	Child swallows the					

stand fin or an LED knocked off | 3 | 2 | 6 | Not a toy, not marketed to children; warning on the box: "Not suitable for children under 3 years – small parts"; LEDs are reflow-soldered | 4 | | Eye (photobiological) | Staring at the LEDs from close up | 1 | 3 | 3 | ≈ 2 lm total, 2700 K, 1.1–8.7 mA per LED: Exempt Group per EN 62471 by inspection | 1 | | Photosensitive epilepsy | Flashing animation | 3 | 1 | 3 | No animation or face flashes above 3 Hz with large-area contrast (Fireworks bursts, sneeze sparkle and sparkle transitions are local or < 1 s; the fastest full-area change is a 0.9 s dissolve); design rule for all future content | 2 | | Tip-over | Tree knocked over on a table | 1 | 3 | 3 | Stand fin 36 mm; low mass; no hazard beyond the fall | 1 | | Chemical | Lead in solder / finish | 2 | 1 | 2 | RoHS: lead-free solder paste and LF-HASL | 1 | | Misuse outdoors | Used in rain | 2 | 2 | 4 | "Indoor use only" on product + box; 5 V cannot cause a shock | 2 |

Conclusion: no residual risk ≥ 8. The product is safe under normal and reasonably foreseeable use, with the listed warnings.

Production control (Module A — internal production control)

1. **Incoming:** JLC/CM supplies the certificate of conformity for the PCB (LF-HASL, UL94 V-0) and the reel labels (lot numbers) for the LEDs and MCU.
2. **Programming:** pogo jig on TP1–TP3, TP5, TP6 (SWD, connect-under-reset), flash, read-back verify.
3. **Functional test (8 s, automatic after flashing):** the test firmware lights each of the 16 scan phases in turn. The operator sees a "rolling" pattern, so a dead line shows as a missing diagonal. Then all 240 LEDs light dimly for a visual check of every LED. Then a touch test on the star.
4. **Batch numbering:** `YYWW-L` (year, week, line) written on the back silkscreen field with a paint marker, or laser-printed on the box label. The batch record stores firmware version, LED reel lot and test yield.
5. **Change control:** any change of the LED part, MCU, PCB outline, clock speed or scan timing triggers an EMC pre-scan and a review of whether the test report still covers the product.